



An Undergraduate's Guide to the Rubens Tube: basic theory, construction, research, and outreach

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The Rubens tube is a popular demonstration to teach students and the public about waves. To operate, a speaker is used to form pressure waves within natural gas in a tube, producing flames with varying heights. A Rubens tube was constructed for use in outreach and to measure the speed of sound in the propane gas used. The speed of sound was measured to be 365 m/s, a 41% difference from the value quoted in the literature. The tube was successful at encouraging curiosity among physics students and the general population alike.

I. INTRODUCTION

The Rubens tube, or flame tube, was developed by Heinrich Leopold Rubens (1865-1922) and Otto Krigar-Menzel (1861-1930) as a method of visualizing waves. The paper describing it, *Flammenröhre für akustische Beobachtungen*, was published in 1905, and the first documented demonstration using it was two years later when U. Behn cited the paper and gave additional experiments to perform [1]. A precursor to the flame tube was developed by Rubens' doctoral advisor, August Kundt, who used a tube filled with fine powder to provide a similar visualization. Rubens used this design and work by U Behn describing the variation in flames due to ambient pressure to construct his flame tube.

Since its advent, the Rubens tube has been used abundantly in acoustics demonstrations to students and the public alike. It has a good mix of simplicity and excitement to capture an audience's interest and curiosity. The visualization aspect helps us understand waves, their behavior, and their anatomy, both on the educational and research levels. Variations have been made to help further our understanding, including using different fuels, different intensities, and even making a two-dimensional flame table in contrast to the standard tube [2]. Because of these uses, we decided to build our own Rubens tube to better understand and teach the physics at play.

II. THEORY

A Rubens tube is a device that visualizes waves. It consists of a metal tube with holes perforated in a line along one side. Gas is pumped through, and waves are formed in the gas with the help of a membrane on one or both of the ends of the tube (fig. 1). When the tube is ignited, an unequal pressure distribution in the gas creates jets of fire of varying heights along the tube. The Rubens tube relies on standing waves. Unlike a traveling wave,

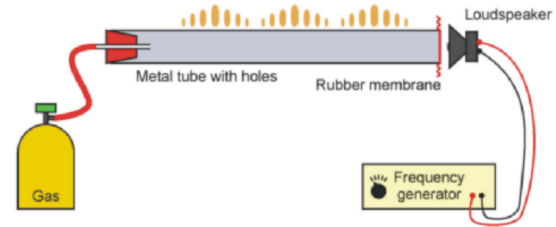


FIG. 1. The standard design of a Rubens tube. Natural gas is pumped into the tube. A speaker plays sound into a membrane on the end, which forms a standing wave within the tube. A line of holes along the top of the tube is ignited, and the standing pressure waves are demonstrated by the varying height of the flames. [3]

where the profile of the wave moving through space, a standing wave resembles a stationary wave that changes in amplitude. The nodes are defined as the points along the wave that always have an amplitude of zero, while the antinodes oscillate between the wave's maximum and minimum. Each complete wave contains 2 nodes and 2 antinodes.

Standing waves can be constructed by superimposing two waves traveling in opposite directions. Standing waves are commonly constructed in a closed-ended tube. Such a tube of length L will have wavelength λ_n that can be modeled by

$$\lambda_n = \frac{2L_{\text{closed}}}{n}, \quad (1)$$

where n is an integer. These instances are referred to as normal modes. The factor of 2 results from the two closed barriers on either end of the tube. The amplitude of a standing wave at these points must be equal to zero, indicating that these points are nodes, and thus viable standing waves must come in multiples of half a wavelength. In these normal modes, a traveling wave in one

direction reflects off the boundary to form a similar wave traveling in the opposite direction. When these waves superimpose, they form a perfect standing wave.

If the medium at the end of the tube is free to vibrate (an “open” end), the picture changes. The flexible membrane of a Rubens tube represents this kind of configuration. At this end, the amplitude of the standing wave must be the maximum, while the other remains a node. The normal mode wavelengths change to

$$\lambda_n = \frac{4L_{\text{open-closed}}}{n}, \quad (2)$$

where n is an odd integer [4].

The normal modes have, thus far, been given in wavelengths. While this is useful to understand the mechanics, it is slightly misleading. To drive waves, a wavelength cannot be directly selected. Instead, the appropriate frequency f must be produced. The speed of wave propagation c is used to convert the two, via the equation

$$c = \lambda f. \quad (3)$$

The speed of sound relies on the medium the wave is propagating through, for reasons beyond the scope of this paper.

In the operation of a Rubens tube, a speaker is used to create pressure waves within the pipe. When the tube is ignited, the unequal pressure distribution in the gas creates jets of fire of varying heights along the tube. The heights of these flames mimic the pressure oscillations within the tube. We now refer to two different waves: the pressure waves in the gas within the tube, and the visual flame waves that represent the pressure waves. Under ideal, high gas pressure operation, the flame maxima occur at the nodes of the standing wave within the tube, where the pressure stays constant. Thus, one complete pressure wave corresponds to two complete visual flame waves. At the flame minima, the pressure wave moves the gas from side to side in the tube rather than letting it escape through the holes. This means less gas is available, and the flames are smaller. The flame wave behavior can be affected by factors such as gas pressure, reversing this effect and causing the flame maxima to occur and the pressure antinodes instead [5]. This occurs due to an effect known as “gulping”. In these cases, The nodes have a lower pressure than their surroundings, so air is drawn through the holes into the tube, limiting the gas supply in these places. A similar effect is observed if the amplitude of the oscillations is adequately high [6].

III. CONSTRUCTION AND OPERATION

As described in Section II, the Rubens tube consists of a perforated steel tube with a flexible membrane on the end. We based our construction on the work of Christopher Sandoval [7]. The tube was to be between 1.5 and 2 m long, with holes every 4 cm except for 10 cm at the

end. These dead spaces at the end are to preserve the integrity of the membrane and gas inlet. The holes were to be drilled with a 2mm drill bit. The flexible membrane was to be a section of a balloon affixed with a hose clamp. Upon the other end of the tube, a hole would be drilled into a pipe cap, and an adapter flare would be fit inside. The adapter flare and pipe cap would be sealed using epoxy.

Tools and materials were limited, so an unconventional and inexact approach was given to actual construction. Using a hacksaw, a 5.7 cm (2.25 inch) diameter steel duct pipe was cut down to 1.37 m (4 feet 6 inches). This cut end had a deviation of 1cm. As planned, holes were drilled through one side of the pipe at a distance of about 4 cm apart using a drill press. The difficulty was encountered with the drill bits breaking. Because of this, three different-sized bits were needed, deviating by 0.0127 mm. The dead space on the ends was 10 cm and 13 cm. To construct the gas inlet, a 1.27 cm (0.5 inch) drill bit was used to drill a hole through a rounded pipe cap. The whole was further increased using a steel file until it was large enough to insert the adapter flare from the inside. The joint was sealed with epoxy, and the pipe cap was similarly sealed onto the steel tube. These were then left to set for 24 hours. The other end of the pipe was sanded down to remove rough edges. The flat of a vinyl glove was attached as a membrane using a hose clamp. However, this setup proved faulty, and the glove was replaced with parafilm to provide a tight seal. The finished tube is shown by fig. 2, and it took an estimated 8 hours over three days to complete.

There are many options for the stand. We constructed two legs out of wooden boards, as is visible in fig. 2. This provided support for the tube at about waist height and ensured the tube would not roll onto its side. The wood was cut using a hand saw and constructed using an electric drill. Unfortunately, this method resulted in uneven cuts, so the setup is not the most sturdy stand, but it functioned well if the tube was not jostled.

We mounted the Rubens tube on a sturdy wooden stand for operation, and a gas supply tank was connected to the adapter flare through a regulator. For our purposes, propane gas was used. The gas was turned on and allowed to fill the tube, at which point the emission holes were lit. The holes were far enough apart that they could not light off one another, so each hole had to be lit manually. A speaker was held to the membrane and tones were produced by a frequency generator. When the tone had a frequency that would create a standing wave in the pipe (as explained in Section II), distinct wavelike features were displayed by the flames (fig. 3). Otherwise, the flames had a low, equal height (fig. 4). Using our wave generator and speaker, we were able to find frequencies that produced 2 through 6 maxima along the tube. On occasion, 7 maxima were observed, however these were not as distinct and not easily reproducible. We frequently needed to relight emission holes, as at the flame minima, the fire would often die out completely.



FIG. 2. The completed Rubens tube. One end is a parafilm membrane, while the gas is pumped into an adapter at the other. A rudimentary wooden stand holds the tube. The emission holes are small enough to not be visible in this image.



FIG. 3. The Rubens tube when containing a standing wave. Note the visible waves in the flames.

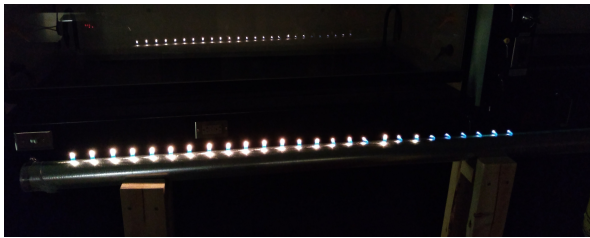


FIG. 4. The Rubens tube when driving at a frequency that does not produce a standing wave. Note the uniformity of the flames along the length of the pipe.

IV. EXPERIMENT: SPEED OF SOUND

An experiment was performed to find the speed of sound in propane fuel. As discussed in Section II, the wavelength of the pressure wave in the tube depends on the speed of sound through the material, in this case propane gas. The speed of sound c must be calculated from the audio frequency f and the wavelength λ using eq. (3).

To perform the experiment, the Rubens tube was set up, connected to the propane supply, and lit. A speaker was hooked up to a frequency generator and placed at the membrane end of the tube to drive the system. The room lighting was adjusted to ensure the flames were visible as well as any measuring tools. The frequency of the speaker was varied to produce a strong wavelike pattern in the flames. The frequency was noted, the number of waves in the tube was counted, and the wavelength was measured using a meter stick. It is important to note that, as explained in Section II, A complete visual wave in the fire corresponds to half of a pressure wave. The counts and measurements were adjusted in the data analysis to account for this. To provide a fit line and value of the speed of sound, eq. (3) was rearranged into

$$\lambda = \frac{c}{f} + b, \quad (4)$$

where c is the speed of sound and b is a correctional term.

An initial analysis was made using the wavelength calculated from the length of the tube l and the number of waves n by taking

$$\lambda = \frac{l}{n}. \quad (5)$$

Using the measured wavelength, the speed of sound was found to be $c = 365 \pm 1 \frac{\text{m}}{\text{s}}$ (fig. 5). This speed, calculated by the fit curve in eq. (4), is higher than the popular value of $258 \frac{\text{m}}{\text{s}}$, but closer to the speed of sound in air, $343 \frac{\text{m}}{\text{s}}$.

With our limited data and testing, the source of this deviation is unknown, however there are multiple possibilities. The gas inside the tube was not well monitored nor controlled, and it is highly likely that there was a mixture of gasses as opposed to pure propane. Additionally, the speed of sound varies not only with the composition of the medium but the temperature as well. This was not considered in testing, and the temperature of the tube was not measured, nor can it easily be assumed. Further work should mount a thermal sensor within the tube to measure gas temperature to obtain a more accurate model for the wave behavior.

The calculation of uncertainty is detailed in Appendix B.

V. OUTREACH

Outreach has two main goals: to educate and to engage. Of these, many would agree fostering engagement

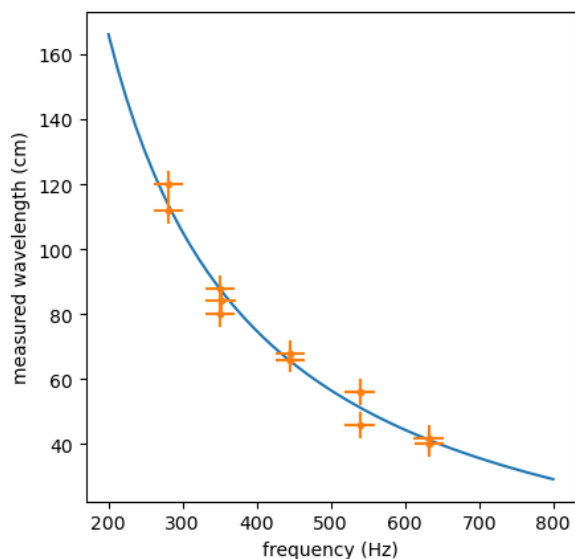


FIG. 5. The trend of frequency versus wavelength over a constant length. The coefficient of the fit provides a value for the speed of sound in the fuel, $c = 365 \pm 1 \frac{\text{m}}{\text{s}}$.

is more important, as that is what drives people to pursue further understanding. The Rubens tube is useful for these purposes thanks to its performative operation and the varying depths at which it can be understood. Many people of all ages are entertained and interested by fire. This interest is hardwired into our brains as an evolutionary mechanism to learn to control fire [8]. This engagement can be demonstrated by an experience with this particular tube, whereupon testing it, nearby classes took a recess to investigate and learn more. Thanks to this inherent interest, the audience is more engaged and curious as to how the flames act the way they do when a standing pressure wave is introduced. In addition, the tube's simplicity welcomes theorizing, furthering engagement. Even if the demonstration does not provide much in the way of education, the audience should walk away with a greater appreciation and interest in science.

The Rubens tube demonstrates a variety of phenomena at different levels, making it versatile for demonstrations and education. Depending on the knowledge and interest of the audience, discussion can include topics from basic wave behavior to pressure waves to even the complexities of gas flow. Care must be taken by the demonstrator though, as if they properly nurture engagement and curiosity, the audience may ask questions that are more in-depth than they are prepared to answer.

VI. CONCLUSION

The Rubens tube is a simple device that visualizes standing pressure waves using flames of varying heights. While construction is simple, care should be taken to ensure the best product. We highly recommend ensur-

ing the possession and location of all necessary tools and materials prior to construction, especially extra drill and driver bits as needed. For the ease of the operator, it is recommended to place the holes close together so that they will automatically ignite from the heat of the others.

From our results, we might conclude that a Rubens tube is not the simplest nor most accurate device for measuring the speed of sound, although a more careful consideration of gas parameters may improve this result. Future measurements should include the temperature of the gas inside the tube and, if possible, an analysis on the mixture of the gas. Independently, the Rubens tube possesses a certain accessibility that makes it particularly valuable for demonstrations.

Appendix A: Construction and Operation Materials

The following materials were consumed in the construction of the tube:

- steel duct tube (1.37 m length, 5.7 cm diameter)
- steel tube end cap
- adapter flare
- steel weld epoxy
- parafilm
- hose clamp

The following tools were used in tube construction:

- drill press
- 0.08 inch drill bit (extras as needed in the event of breakage)
- 0.5 inch drill bit
- hacksaw
- sandpaper
- 3 clamps
- metal file

The following materials were used in construction of the stand:

- 16 screws
- 2 wooden beams of dimensions 1.8 m $\times$ 5.1 cm $\times$ 10.2 cm (6 ft $\times$ 2 in $\times$ 4 in)

The following tools were used in stand construction:

- handsaw
- electric screwdriver
- driver bit

- drill bit
- 2 clamps

The following materials are required for tube operation:

- Rubens Tube
- stand
- gas flow regulator
- fuel tank
- speaker
- frequency generator

Appendix B: Uncertainty

The uncertainty in measured wavelength was taken to be 4cm, the distance between holes. Uncertainty in num-

ber of waves was the square root of the count. Error of audio frequency was 20 Hz, the resolution of the measuring device. All calculated errors followed the form

$$\text{if } y = f(x), \sigma_y = \frac{df}{dx} \sigma_x,$$

where σ denotes uncertainty.

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